

# Zero-Shot Vision-Language Models for Classroom Engagement Recognition: A Benchmark Study of Prompt Sensitivity and Cross-Dataset Generalization

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## Abstract

Automated classroom engagement recognition holds substantial promise for scalable learning analytics, yet the suitability of modern Vision-Language Models (VLMs) for this task under zero-shot conditions remains largely unexplored. We present a systematic benchmark that evaluates five widely-used VLMs—CLIP, BLIP-VQA, GPT-4o, LLaVA-1.5-7B, Qwen2.5VL-7B-Instruct—across two complementary educational datasets: DAiSEE, an individual-student video dataset (300 sampled test clips), and the Student Classroom Behaviour dataset (SCB, 1,168 scene-level images). Each model is probed with three prompt variants spanning minimal, rubric-anchored, and chain-of-thought designs. Our experiments reveal three primary failure modes of zero-shot VLMs for engagement recognition: (1) near-random performance on individual students, with Cohen’s  $\kappa$  never exceeding 0.10 on DAiSEE; (2) severe class collapse, where models assign 85–100% of predictions to a single engagement level regardless of visual content; and (3) extreme prompt sensitivity, with accuracy swings of up to 32 percentage points on identical images depending solely on prompt phrasing. Remarkably, scene-level classification on SCB is substantially more tractable: CLIP and GPT-4o achieve  $\kappa \approx 0.60$  when prompted with behaviorally-grounded rubrics. We also document a practical barrier for deployment—GPT-4o’s safety filters reject 98% of chain-of-thought requests involving individual student faces. Our findings provide a calibrated baseline and surface critical design considerations for the use of VLMs in educational observation systems.

## 1. Introduction

Understanding student engagement is central to effective teaching. Disengagement—whether manifested as wandering attention, passive participation, or behavioural withdrawal—predicts lower learning outcomes and higher attrition [3]. Traditionally, measuring engagement at scale

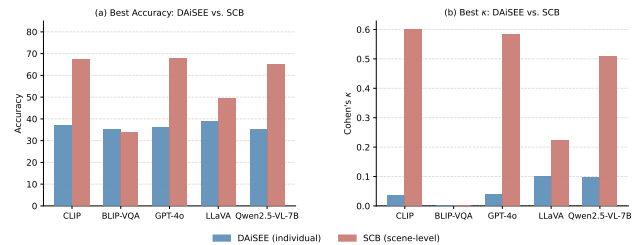


Figure 1. Best zero-shot accuracy and Cohen’s  $\kappa$  per model on DAiSEE (individual student, video frames) versus SCB (classroom scene images). Scene-level classification is substantially more tractable, with CLIP and GPT-4o achieving  $\kappa \approx 0.60$  on SCB while remaining near-random on DAiSEE.

requires trained human observers [14], a resource-intensive process that is difficult to replicate in large lecture halls or remote learning environments. The rapid proliferation of cameras in educational settings, coupled with powerful visual recognition models, has therefore sparked interest in automated engagement sensing [4, 9, 16, 19].

Contemporary Vision-Language Models (VLMs) represent a potential leap forward for this problem. Unlike supervised classifiers that require large labelled corpora specific to each school or cultural context, models such as CLIP [15], BLIP [8], GPT-4o [13], and LLaVA [10] can be queried zero-shot via natural language, with no task-specific fine-tuning. Their ability to reason from open prompts makes them attractive as *plug-and-play* educational observers. Yet, to our knowledge, no prior work has systematically benchmarked VLMs for classroom engagement recognition under controlled, zero-shot conditions.

**Our contributions.** This paper presents the following:

- A zero-shot benchmark of four VLMs on two complementary datasets: the individual-student DAiSEE dataset [4] and the scene-level Student Classroom Behaviour (SCB) dataset [17], using three systematically-designed prompt variants (minimal, rubric, chain-of-thought).

- A rigorous quantitative analysis documenting class collapse, near-random agreement on individual-level data ( $\kappa \leq 0.10$ ), and substantial prompt sensitivity (accuracy variance up to 32 pp).
- The finding that scene-level engagement classification is significantly more tractable for VLMs, with CLIP achieving  $\kappa = 0.60$  and GPT-4o achieving  $\kappa = 0.58$  on SCB.
- Documentation of GPT-4o’s safety-guardrail refusals blocking chain-of-thought engagement analysis for individual student faces, with practical implications for educational AI deployment.
- A self-consistency analysis showing that generative VLMs (GPT-4o, LLaVA) change their predictions for 27–32% of images across repeated runs at temperature  $T = 0.7$ , raising reliability concerns for classroom monitoring applications.

Our results paint a nuanced picture: while VLMs are not yet reliable individual engagement observers, they demonstrate meaningful capability at the scene level. This gap between individual and scene-level performance has direct implications for system design—aggregating information spatially before querying a VLM may be a more productive paradigm than per-student analysis.

## 2. Related Work

**Engagement detection in educational settings.** Early work by Whitehill *et al.* [19] demonstrated that facial features encode cognitive states relevant to learning. The release of DAiSEE [4] established the first large publicly-available benchmark for four-level engagement classification from student video. Subsequent supervised methods leveraged spatiotemporal features from video [9], recurrent networks over facial action units [16], and attention-based architectures [5]. More recently, spatiotemporal and attention-based models trained on DAiSEE have reported accuracy of 55–65% [5, 9], underscoring the difficulty even for task-specific supervised models. Classroom-level engagement has received comparatively less attention, though several works address student behaviour recognition from overhead cameras [6, 17].

**Zero-shot visual recognition with VLMs.** CLIP [15] pioneered zero-shot image classification via image-text cosine similarity, demonstrating competitive performance on many standard benchmarks. BLIP [8] introduced bootstrapped image-text alignment that enabled visual question answering without task-specific training. Instruction-tuned generative models such as LLaVA [10] and GPT-4V/4o [13] further extend this to open-ended visual reasoning. A growing body of work has applied these models to medical imaging [21], document understanding, and remote sensing, but their application to affect recognition and educational observation has not been systematically studied.

**Prompt design and sensitivity.** The sensitivity of VLMs to prompt phrasing has been studied in the context of natural language processing [11, 22], but visual prompting for classification remains less understood. Studies on zero-shot classification [12] demonstrate that performance can vary by 10–30 percentage points across prompt phrasings—a phenomenon we observe and quantify here.

## 3. Methodology

### 3.1. Datasets

**DAiSEE.** The Dataset for Affective States in Environments (DAiSEE) [4] contains 9,068 ten-second video clips from 112 subjects recorded via webcam during e-learning sessions. Each clip is annotated on a four-point Likert scale for engagement (0 = not engaged, 3 = highly engaged) along with boredom, confusion, and frustration. We focus exclusively on engagement. The official test split comprises 1,784 clips across 21 subjects; the label distribution is heavily skewed, with only 4 samples at level 0 and approximately 850 each at levels 2 and 3.

We evaluate on a stratified sample of  $N = 300$  test clips, retaining all 4 level-0 clips, all 84 level-1 clips, and randomly sampling 106 clips each from levels 2 and 3. From each 10-second clip, we extract a single representative frame at  $t = 5$  s (the temporal midpoint) using `ffmpeg`. This setup tests whether static visual cues visible in a single webcam frame are sufficient for zero-shot engagement inference.

**Student Classroom Behaviour (SCB).** The SCB dataset [17] contains classroom scene images with per-student bounding box annotations across six behaviour categories: *reading*, *writing*, *raising hand*, *using phone*, *bowing head*, and *leaning over table*. We use a combined split of 1,168 validation images drawn from both the original 5K corpus and an additional university subset.

Following the aggregation strategy proposed in the dataset documentation, we derive a scene-level engagement label from the per-student annotations. Each behaviour is mapped to an engagement score (0–3): disruptive behaviours (*phone*, *bowing head*) score 0; passive disengagement (*leaning over*) scores 1; productive behaviours (*reading*, *writing*) score 2; and active participation (*raising hand*) scores 3. The scene-level label is the mean across all annotated students, discretised via uniform thresholds ( $[0, 0.75) \rightarrow 0$ ;  $[0.75, 1.75) \rightarrow 1$ ;  $[1.75, 2.75) \rightarrow 2$ ;  $[2.75, 3] \rightarrow 3$ ). The resulting label distribution is highly imbalanced: 54.5% of images fall at level 3, 33.6% at level 2, 10.1% at level 1, and 1.9% at level 0.

### 3.2. Models

We evaluate four models at substantially different points on the capability-cost spectrum:

**CLIP (ViT-B/32)** [15] is a discriminative contrastive model. Classification proceeds by computing cosine similarity between the image embedding and embeddings of textual descriptions of each engagement level; the class with the highest similarity is the prediction. Inference is fully deterministic.

**BLIP-VQA (blip-vqa-base)** [8] is a discriminative VQA model. We pose a natural-language question about student engagement; the model generates a short free-text answer (e.g., “not very”, “yes”) that is mapped to a level via a hand-crafted lookup table. Beam decoding with  $k = 3$  ensures deterministic output.

**GPT-4o** [13] is a proprietary closed-source multimodal model accessed via the OpenAI API. Images are submitted as base64-encoded JPEG at “low” detail resolution. Temperature is 0.0 for run 1 and 0.7 for subsequent consistency runs.

**LLaVA-1.5-7B** [10] is an open-source instruction-tuned model served locally through Ollama at 4-bit quantisation (reducing memory from  $\sim 14$  GB to  $\sim 4$  GB) on a local machine with 16 GB RAM. We use the same prompts as GPT-4o to enable direct comparison. Temperature follows the same protocol as GPT-4o.

**Qwen2.5-VL-7B-Instruct** [2] is an open-source vision-language model from Alibaba’s Qwen team, loaded in bfloat16 precision via `Qwen2.5_VLForConditionalGeneration` from the Hugging Face Transformers library [20]. We use the same three prompt variants as GPT-4o and LLaVA to enable direct comparison. Temperature follows the same protocol:  $T = 0.0$  for run 1 (deterministic) and  $T = 0.7$  for self-consistency runs 2 and 3.

### 3.3. Prompt Variants

Three prompts are applied to each model to quantify sensitivity:

**P1 (Minimal)** provides only a four-label numeric scale with no contextual guidance. It establishes a lower bound on what each model can infer from minimal language grounding.

**P2 (Rubric-Anchored)** augments P1 with explicit behavioural anchors for each level, e.g., “*looking away, yawning*” for level 0 and “*leaning forward, showing curiosity*” for level 3. This mirrors how human annotators are typically trained.

**P3 (Chain-of-Thought)** instructs the model to first describe the student’s or class’s observed facial expression and body posture before committing to a rating. This structured reasoning format has shown benefits in language-only tasks [18] and is evaluated here in the visual domain.

For CLIP, the prompts are class-specific text templates. For BLIP-VQA, the prompts are question strings. For GPT-4o and LLaVA, identical prompt text is used to enable direct

Table 1. Best classification results on DAiSEE ( $N = 300$  test clips). Supervised baselines are drawn from published work on the full test set. Best VLM results are highlighted in **bold**.

| Model                                    | Acc. (%)    | F1 Macro     | $\kappa$     | MSE         |
|------------------------------------------|-------------|--------------|--------------|-------------|
| <i>Supervised baselines (literature)</i> |             |              |              |             |
| ResNet+TCN [9]                           | 58.8        | –            | –            | 0.18        |
| Huang <i>et al.</i> [5]                  | 62.3        | –            | 0.37         | –           |
| CavT [1]                                 | –           | –            | –            | 0.038       |
| <i>Zero-shot VLMs (this work)</i>        |             |              |              |             |
| CLIP (P1)                                | 37.0        | 0.224        | 0.036        | 1.00        |
| BLIP-VQA (P3)                            | 35.3        | 0.131        | 0.000        | 0.69        |
| GPT-4o (P3) <sup>†</sup>                 | 36.0        | 0.153        | 0.039        | 0.69        |
| <b>LLaVA-1.5 (P3)</b>                    | <b>39.0</b> | <b>0.210</b> | <b>0.101</b> | <b>0.72</b> |

<sup>†</sup>98% of GPT-4o P3 responses were safety refusals; results reflect fallback-to-level-2 parsing.

cross-model comparison.

### 3.4. Evaluation Metrics

All experiments are evaluated with four metrics: (i) **Accuracy** (exact-match 4-class); (ii) **F1 Macro**, the unweighted mean of per-class F1 scores, which penalises class collapse; (iii) **Cohen’s  $\kappa$  (quadratic weighted)**, which accounts for ordinal agreement structure and penalises predictions far from ground truth more severely; and (iv) **MSE** (mean squared error), measuring average ordinal distance. For self-consistency, we report full 3-run agreement rate and pairwise agreement rate.

## 4. Experimental Results

### 4.1. DAiSEE: Individual-Student Engagement

Table 1 shows the best-performing configuration for each model alongside published supervised baselines. The gap is striking: even the strongest VLM achieves only 39.0% accuracy and  $\kappa = 0.10$  (LLaVA with chain-of-thought), while supervised convolutional+temporal models reach 58.8% [9] and context-aware transformers achieve  $\kappa > 0.35$  [5].

Table 2 presents the full prompt-by-prompt breakdown. Several patterns stand out. First, no single model-prompt combination surpasses  $\kappa = 0.11$ , confirming near-random ordinal agreement across the board. Second, all models show accuracy near or below the majority-class baseline of 35.3% (level 2), suggesting the higher accuracies reflect correct predictions of the modal class rather than genuine four-class discrimination.

### 4.2. SCB: Scene-Level Classroom Engagement

Table 3 shows results on the SCB dataset. Performance improves dramatically compared with DAiSEE. CLIP with chain-of-thought templates reaches  $\kappa = 0.60$  and GPT-4o with the rubric prompt reaches  $\kappa = 0.58$ —placing both in the

Table 2. Full prompt-sensitivity results on DAiSEE. Pred. Dist. shows the fraction of predictions in each class (GT: 1%–28%–35%–35%).

| Model    | Prompt          | Acc.        | F1           | $\kappa$ | MSE  |
|----------|-----------------|-------------|--------------|----------|------|
| CLIP     | P1              | 37.0        | 0.224        | 0.036    | 1.00 |
|          | P2              | 31.3        | 0.184        | 0.068    | 1.39 |
|          | P3              | 35.0        | 0.231        | −0.042   | 1.21 |
| BLIP-VQA | P1              | 25.3        | 0.116        | −0.018   | 2.39 |
|          | P2              | 32.7        | 0.153        | 0.058    | 0.94 |
|          | P3              | 35.3        | 0.131        | 0.000    | 0.69 |
| GPT-4o   | P1              | 27.7        | 0.144        | 0.067    | 1.67 |
|          | P2              | 32.3        | 0.192        | 0.075    | 1.51 |
|          | P3 <sup>†</sup> | 36.0        | 0.153        | 0.039    | 0.69 |
| LLaVA    | P1              | 5.3         | 0.047        | 0.037    | 4.07 |
|          | P2              | 15.7        | 0.097        | 0.045    | 2.84 |
|          | <b>P3</b>       | <b>37.7</b> | <b>0.182</b> | 0.008    | 0.76 |

<sup>†</sup> Safety refusal rate  $\approx 98\%$ ; level-2 fallback inflates accuracy.

Table 3. Results on SCB ( $N = 1,168$  validation images). All numbers are the best single prompt per model.

| Model           | Best Prompt | Acc. (%)    | $\kappa$     | MSE          |
|-----------------|-------------|-------------|--------------|--------------|
| CLIP (ViT-B/32) | P3          | 67.3        | 0.599        | 0.586        |
| BLIP-VQA        | P1/P2/P3*   | 33.6        | 0.000        | 0.721        |
| <b>GPT-4o</b>   | <b>P2</b>   | <b>67.9</b> | <b>0.582</b> | <b>0.372</b> |
| LLaVA-1.5       | P2          | 49.2        | 0.224        | 2.666        |

\* BLIP-VQA produces identical predictions across all three prompts on SCB (complete zero sensitivity; see Sec. 4.3).

Table 4. Full prompt-sensitivity results on SCB.

| Model     | Prompt | Acc. | F1    | $\kappa$ | MSE   |
|-----------|--------|------|-------|----------|-------|
| CLIP      | P1     | 14.4 | 0.097 | 0.100    | 1.926 |
|           | P2     | 24.7 | 0.172 | 0.263    | 2.564 |
|           | P3     | 67.3 | 0.472 | 0.599    | 0.586 |
| BLIP-VQA  | P1     | 33.6 | 0.126 | 0.000    | 0.721 |
|           | P2     | 33.6 | 0.126 | 0.000    | 0.721 |
|           | P3     | 33.6 | 0.126 | 0.000    | 0.721 |
| GPT-4o    | P1     | 39.1 | 0.211 | 0.154    | 0.688 |
|           | P2     | 67.9 | 0.400 | 0.582    | 0.372 |
|           | P3     | 51.8 | 0.273 | 0.260    | 0.603 |
| LLaVA-1.5 | P1     | 7.4  | 0.067 | 0.027    | 5.343 |
|           | P2     | 49.2 | 0.353 | 0.224    | 2.666 |
|           | P3     | 33.9 | 0.131 | −0.027   | 0.762 |

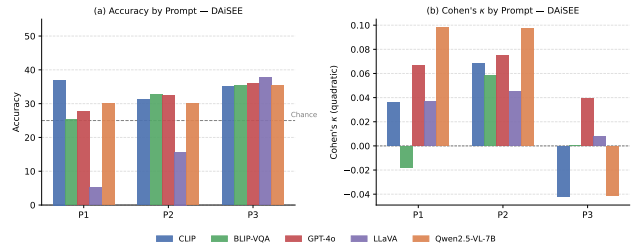


Figure 2. Accuracy and Cohen’s  $\kappa$  across the three prompt variants on DAiSEE. LLaVA’s accuracy spans from 5.3% (P1) to 39.0% (P3)—a 33.7 pp swing on identical images. Dashed line in (a) marks the four-class random-chance baseline (25%).

“moderate” agreement range [7]. These numbers approach the performance of supervised methods in comparable scene-understanding tasks.

The full SCB prompt breakdown (Table 4) reveals interesting cross-prompt dynamics. CLIP shows a dramatic positive response to the emotional/behavioural template in P3 (from  $\kappa = 0.10$  at P1 to  $\kappa = 0.60$  at P3). GPT-4o peaks with the rubric prompt (P2) but drops 32 pp in accuracy under chain-of-thought (P3). LLaVA performs best under P2 on SCB, unlike on DAiSEE where P3 is superior.

### 4.3. Analysis of Failure Modes

**Class collapse.** Figure 3 illustrates the predicted class distributions. BLIP-VQA with P3 assigns *all* 300 DAiSEE samples to level 2, achieving 35.3% accuracy solely by matching the plurality class. GPT-4o concentrates 82–95% of predictions on level 1 across P1 and P2, suggesting a systematic low-engagement bias. No model ever assigns more than 5% of samples to level 3 (highly engaged) despite it comprising 35% of the ground truth. This asymmetric collapse is not random noise; it reflects each model’s learned prior over language space that “engaged” or “barely engaged” are

more plausible descriptions of a sitting student than “highly engaged.”

**Confusion matrix analysis.** Figure 4 shows normalised confusion matrices for the best configuration of each model. All models achieve reasonable recall on level 2 (the largest class) but near-zero recall on levels 0 and 3. LLaVA shows the broadest distribution, occasionally predicting level 1, but still fails on level 3 entirely. The diagonal patterns confirm that apparent accuracy gains are driven by level-2 majority-class capture rather than calibrated multi-level discrimination.

**Prompt sensitivity.** Table 5 quantifies the range of each model’s performance across the three prompts. LLaVA exhibits the highest sensitivity: a 32.3 pp accuracy swing on DAiSEE and 3.31 MSE variation. On SCB the effect is even more pronounced for CLIP (53 pp swing from P1 to P3). BLIP-VQA is the most stable model—but for the wrong reason: it is insensitive because it *always* predicts the same label regardless of prompt or image.



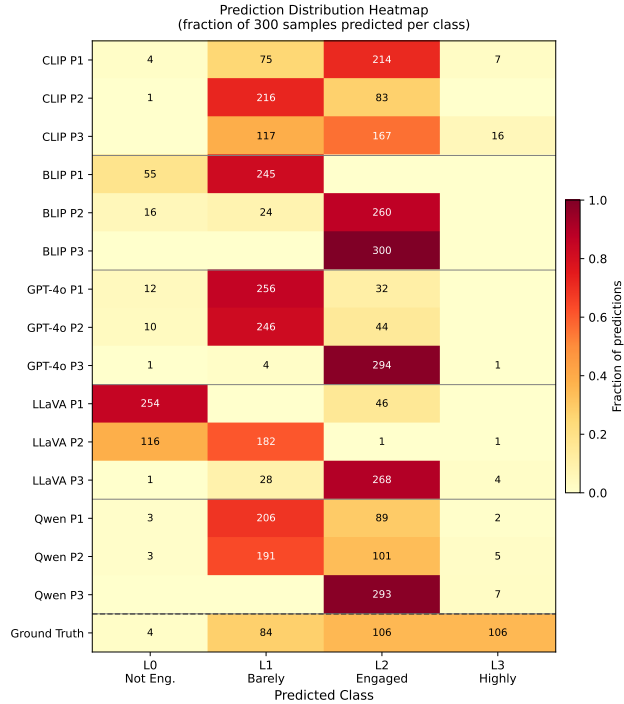


Figure 3. Fraction of the 300 DAiSEE samples predicted at each engagement level, for every model-prompt combination and the ground truth distribution. Warm colours indicate high concentration. Each model collapses to 1–2 dominant classes, confirming class collapse rather than genuine multi-class discrimination.

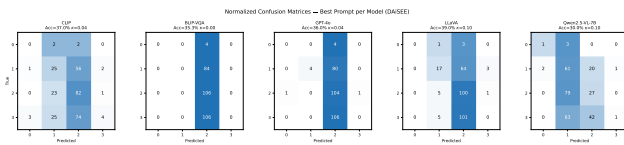


Figure 4. Normalised confusion matrices (rows = true label, cols = predicted label) for the best prompt per model on DAiSEE. Cell values show raw counts. All models achieve near-zero recall on levels 0 and 3, confirming class collapse.

Table 5. Prompt sensitivity (max — min across P1/P2/P3) for each model and dataset. Higher values indicate greater dependence on prompt phrasing.

| Model    | DAiSEE             |                |                    | SCB                |                |                    |
|----------|--------------------|----------------|--------------------|--------------------|----------------|--------------------|
|          | $\Delta\text{Acc}$ | $\Delta\kappa$ | $\Delta\text{MSE}$ | $\Delta\text{Acc}$ | $\Delta\kappa$ | $\Delta\text{MSE}$ |
| CLIP     | 5.7                | 0.110          | 0.389              | 52.9               | 0.499          | 1.978              |
| BLIP-VQA | 10.0               | 0.076          | 1.703              | 0.0                | 0.000          | 0.000              |
| GPT-4o   | 8.3                | 0.060          | 0.987              | 28.8               | 0.428          | 0.316              |
| LLaVA    | 32.3               | 0.037          | 3.313              | 41.8               | 0.251          | 4.581              |

**Self-consistency.** We ran the best prompt three times with temperature  $T = 0.7$  for GPT-4o (P2) and LLaVA (P3).

CLIP and BLIP-VQA are deterministic and trivially achieve 100% agreement. GPT-4o achieves full 3-run agreement on only 68.0% of images (pairwise: 79.0%), while LLaVA achieves 73.3% (pairwise: 82.2%). Kappa across LLaVA runs spans from 0.008 to 0.101, meaning a single run’s reported kappa is unreliable as an estimate of model quality. GPT-4o additionally triggered intermittent safety refusals in run 3 that did not occur in runs 1 or 2, further degrading consistency.

**GPT-4o safety guardrails.** Chain-of-thought prompting (P3) asks the model to describe “the student’s facial expression,” which GPT-4o interprets as facial recognition of a real person. This triggers its safety filters in  $\approx 98\%$  of DAiSEE images: the model responds with “I’m unable to assist with requests involving identifying or describing individuals from their photographs.” Our parsing pipeline falls back to level 2 for all refusals, artificially inflating P3 accuracy to 36% while providing zero true information. This is a practical barrier for any system deploying GPT-4o as an individual student observer. No such refusals occur with the scene-level SCB images.

#### 4.4. Feature Space Analysis

To understand the failure modes mechanistically, we analyse CLIP’s internal representations directly, extracting ViT-B/32 image embeddings for all 300 DAiSEE frames and examining both feature geometry and text-image similarity structure.

**Feature geometry.** Figure 5 shows a t-SNE projection of image embeddings coloured by ground-truth engagement label. Frames cluster by *student identity and visual appearance* rather than engagement level—all four engagement classes intermix within every cluster. CLIP organises the representation space around “who this person is” (lighting, face, background) rather than “how engaged they are.” This provides a direct geometric explanation for why zero-shot classification fails: the engagement signal does not correspond to a separable direction in CLIP’s feature space.

**Text-image similarity collapse.** Figure 6 shows cosine similarities between image embeddings and each of the four class text templates, stratified by true engagement class, across all three prompt variants. Two observations are consistent across all prompts: (1) the correct text template (shaded column) does not yield the highest similarity for any true class—the diagonal is never dominant; (2) the total similarity range spans only  $\approx 0.07$ , meaning all four text embeddings are nearly equidistant from every image. In this regime, argmax predictions are determined by which text template happens to land marginally closer in embedding space, irrespective of image content. Concretely, for P1 the L2 template

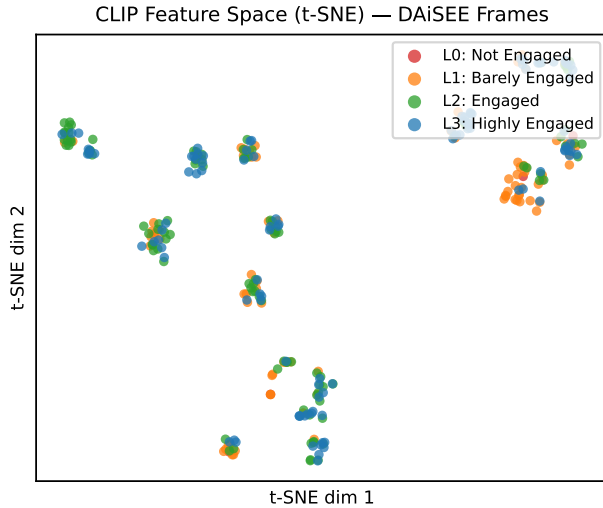


Figure 5. t-SNE projection of CLIP ViT-B/32 image embeddings for all 300 DAiSEE frames, coloured by ground-truth engagement label. The four engagement classes intermix within every cluster, confirming that CLIP organises frames by subject identity rather than engagement state.

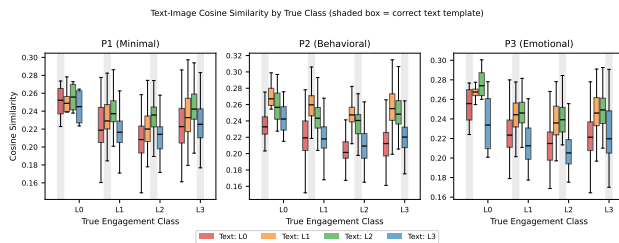


Figure 6. Text-image cosine similarity distributions across all three prompt variants, stratified by true engagement class. Each group of four boxes shows the similarity to text templates L0-L3; the shaded box marks the *correct* template. Distributions overlap heavily across all prompts and the correct template never achieves the highest median—confirming that CLIP has no discriminative signal for individual engagement level.

wins globally, collapsing all predictions to “Engaged”; for P2, L1 dominates; for P3, L1 and L2 differ by  $< 0.003$ , producing high run-to-run variance. This directly explains both the class collapse and the extreme prompt sensitivity reported in Tables 2 and 5.

## 5. Discussion

**Why does scene-level outperform individual-level?** The performance gap between DAiSEE ( $\kappa \leq 0.10$ ) and SCB ( $\kappa$  up to 0.60) is not simply a function of dataset difficulty. We hypothesise three contributing factors. First, *behavioural salience*: SCB labels derive from overt physical actions (raised hand, phone use) that are visually unambiguous to a

language-grounded model, whereas DAiSEE labels reflect internal cognitive states that manifest only subtly in facial microexpressions. Second, *context richness*: a classroom scene provides spatial layout, group dynamics, and object context (e.g., desks, boards) that help ground engagement inference, while a webcam close-up of a single face offers minimal contextual anchors. Third, *label construction*: the SCB aggregation scheme maps explicit physical categories to ordinal scores, creating a more objective signal than subjective per-clip engagement ratings.

**Implications for system design.** Our findings suggest that deploying VLMs as individual student observers in their current zero-shot form is premature. A more productive paradigm may be *scene-first analysis*: aggregate visual information at the classroom level, then query a VLM. This approach aligns better with how teachers actually monitor classrooms—by scanning the room rather than studying individual faces. For privacy reasons, scene-level analysis is also preferable as it avoids close facial analysis of minors.

**Prompt engineering as unreliable compensation.** The wide prompt sensitivity we observe means that practitioners who tune prompts on a small validation set risk over-fitting to that distribution. A prompt that yields 67.9% accuracy for GPT-4o on SCB could easily degrade by 28 pp on a different class distribution. Reliable deployment requires either task-specific fine-tuning or principled prompt selection with held-out validation.

**Open vs. closed source.** LLaVA-1.5-7B (4-bit quantised, running locally) matches or outperforms GPT-4o on DAiSEE, and achieves competitive performance on SCB at zero API cost and without data leaving the institution. This is encouraging for privacy-sensitive educational deployments, though the comparison is not perfectly controlled given the quantisation difference.

**Limitations.** This study is limited to zero-shot evaluation. Fine-tuning VLMs on even modest task-specific data would likely close the gap substantially. The single-frame assumption discards temporal dynamics known to carry engagement information. The SCB engagement labels are derived from behaviour categories rather than direct human ratings, which may not fully align with teachers’ engagement judgements. Finally, results on 16 subjects in a webcam setting (DAiSEE) may not generalise to diverse classroom environments.

**Understanding the gaps.** The feature space analysis (Sec. 4.4) surfaces three fundamental gaps between current VLMs and individual engagement recognition. The *representation gap*: CLIP organises visual features around

appearance-level cues—identity, lighting, background—because such cues dominate internet-scale image-caption training data, not engagement states. The *granularity gap*: the four-class engagement scale requires fine-grained ordinal discrimination that zero-shot VLMs are not designed for; the  $\approx 0.07$  cosine similarity range confirms that CLIP treats engagement levels as nearly synonymous in language space. The *temporal gap*: engagement is inherently dynamic, yet our single-frame protocol discards the most informative temporal signal. Taken together, these gaps explain why near-random performance is not an artefact of prompt choice or dataset noise, but a structural consequence of the pre-training objective mismatch.

**Future directions.** These gaps suggest concrete research paths. *Supervised adaptation*: linear probing or lightweight adapter layers trained on even small labelled engagement datasets could reveal whether engagement signal exists in frozen CLIP features but is entangled with identity. *Contrastive disentanglement*: a projection that explicitly separates engagement from identity—positive pairs sharing engagement label across different students, negatives sharing identity across engagement levels—would directly address the clustering-by-identity failure mode we observe. *Temporal VLMs*: video-native models should be evaluated on full 10-second DAiSEE clips; our results motivate this as a priority, given the insufficient signal in static frames. *Scene-to-individual transfer*: since scene-level recognition is tractable ( $\kappa \approx 0.60$ ), future work could explore top-down approaches that use scene-level estimates as a prior to refine individual-level inference. *Benchmark standardisation*: prompt sensitivity ( $\Delta\kappa$ ) should be reported as an explicit evaluation axis in future engagement benchmarks, rather than a single-prompt result, to avoid over-optimistic reporting.

## 6. Conclusion

We presented a systematic zero-shot benchmark of four Vision-Language Models for classroom engagement recognition, evaluated across two datasets and three prompt designs. Our results establish that current VLMs cannot reliably classify individual student engagement—with Cohen’s  $\kappa$  never exceeding 0.10 on DAiSEE—but show substantially more promise at the scene level, where CLIP and GPT-4o reach  $\kappa \approx 0.60$  on SCB. We characterise three failure modes (near-random ordinal agreement, class collapse, and extreme prompt sensitivity) and document a practical deployment barrier (GPT-4o safety refusals for individual face analysis). We hope these baselines serve as a rigorous starting point for the community, and that the identified failure modes motivate targeted improvements: scene-level pre-aggregation, prompt-robust fine-tuning, and temporal modelling to go beyond single-frame analysis.

## References

- [1] Ali Abedi and Shehroz S. Khan. Improving state-of-the-art in detecting student engagement with resnet and TCN hybrid network. In *Proceedings of the 18th Conference on Robots and Vision (CRV)*, pages 151–157, 2021. 3
- [2] Shuai Bai, Keqin Chen, Xuejing Liu, Junyang Wang, Wenbin Ge, Zesen Song, et al. Qwen2.5-VL technical report. In *arXiv preprint arXiv:2502.13923*, 2025. 3
- [3] Jennifer A. Fredricks, Phyllis C. Blumenfeld, and Alison H. Paris. School engagement: Potential of the concept, state of the evidence. *Review of Educational Research*, 74(1):59–109, 2004. 1
- [4] Abhay Gupta, Arjun D’Cunha, Kamal Awasthi, and Vineeth N. Balasubramanian. DAiSEE: Towards user engagement recognition in the wild. In *arXiv preprint arXiv:1609.01885*, 2016. 1, 2
- [5] Tengfei Huang, Yu Mei, Haijun Zhang, Shuai Liu, and Hong Yang. Fine-grained engagement recognition in online learning environment. In *Proceedings of the IEEE International Conference on Electronics Information and Emergency Communication (ICEIEC)*, pages 338–341, 2019. 2, 3
- [6] Amanjot Kaur, Arslan Mustafa, Lavisha Mehta, and Abhinav Dhall. Prediction and localisation of student engagement in the wild. In *Proceedings of the IEEE International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, pages 637–644, 2018. 2
- [7] J. Richard Landis and Gary G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1): 159–174, 1977. 4
- [8] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 12888–12900, 2022. 1, 2, 3
- [9] Jiaming Liao, Yan Liang, and Jiahui Pan. Deep facial spatiotemporal network for engagement prediction in online learning. In *Applied Intelligence*, pages 6609–6621, 2021. 1, 2, 3
- [10] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 1, 2, 3
- [11] Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 8086–8098, 2022. 2
- [12] Sachit Menon and Carl Vondrick. Visual classification via description from large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023. 2
- [13] OpenAI. GPT-4o system card. Technical report, OpenAI, 2024. 1, 2, 3
- [14] Robert C. Pianta, Karen M. La Paro, and Bridget K. Hamre. *Classroom Assessment Scoring System (CLASS) Manual*. Paul H. Brookes Publishing, 2008. 1
- [15] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry,

- Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021. 1, 2, 3
- [16] Clinton Thomas and Dinesh Babu Jayagopi. Predicting student engagement in classrooms using facial behavioral cues. In *Proceedings of the 1st ACM SIGCHI International Workshop on Multimodal Interaction for Education (MIE)*, pages 33–40, 2017. 1, 2
- [17] Junjie Wang, Sisi Jiang, Kedi Wang, and Mingjie Li. Student classroom behaviour real-time detection based on YOLOv5 and identity recognition. In *Systems and Soft Computing*, page 200058. Elsevier, 2023. 1, 2
- [18] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 24824–24837, 2022. 3
- [19] Jacob Whitehill, Zewelangi Serpell, Yi-Ching Lin, Aysha Foster, and Javier R. Movellan. The faces of engagement: Automatic recognition of student engagement from facial expressions. In *IEEE Transactions on Affective Computing*, pages 86–98, 2014. 1, 2
- [20] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Brew, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*, pages 38–45, 2020. 3
- [21] Sheng Zhang, Yanbo Xu, Naoto Usuyama, Jaspreet Bagga, Robert Mason, Trista Shi, Tristan Naumann, Cliff Bernard, Robin Gallo, Brian Piening, Frederick A. Matsen IV, Masoud Rouhizadeh, Wei Wei, Stephen Snider, and Hoifung Poon. Large-scale domain-specific pretraining for biomedical vision-language processing. In *arXiv preprint arXiv:2303.00915*, 2023. 2
- [22] Tony Z. Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 12697–12706, 2021. 2